Three-Tier Campus Network: Project Overview

Oct 14, 2026 · @Luke

Summary

A three-tier campus network in Packet Tracer covering the intranet: a central router at the core, two building core routers, two Layer 3 distribution switches (the MDFs) and six dual-homed access switches, with a server room, a wireless controller and an eBGP handoff to a simulated MPLS provider. Each tier has one job, and the design shows the hierarchy that Collapse Core condenses into two switches.

- **Scale:** 16 routers and switches (6 routers, 2 Layer 3 switches, 8 Layer 2 switches), 7 servers, a wireless LAN controller and access point, 16 endpoints, and 48 links (42 copper, 6 single-mode fiber).
- **Core:** a central router/firewall node joins the DMZ router, both building cores and the edge router; its firewall role is not configured yet.
- **Distribution and access:** HSRP version 2 on the MDF switches for ten VLANs, and every access switch uplinks to both MDFs.
- **Scope:** the intranet and the edge handoff; internet services and NAT are not part of this stage.

Topology

The design has 46 devices and 48 links (42 copper, 6 single-mode fiber). Traffic flows from the access layer up through two building cores to one central router, which also connects the services segment and the edge.

Tier	Devices	Uplinks
Services	Server-RM switch with DNS, DHCP, web, email, TFTP, AAA and NTP servers	Gi0/1 to DMZ-RTR
DMZ	DMZ-RTR	Fiber Gi0/0/0 to CORE-RTR/FW

Tier	Devices	Uplinks
Core	CORE-RTR/FW (central)	Fiber to BLDG20-CORE (Gi0/1/0), BLDG30-CORE (Gi0/2/0) and EDGE (Gi0/3/0)
Distribution	BLDG20-CORE-RTR with BLDG20-MDF-L3; BLDG30-CORE-RTR with BLDG30-MDF-L3	Each building core Gi0/1 to its MDF Gi0/1; MDFs trunked on Gi0/2
Access	FL1 to FL6 (FL6 is the executive office), WLC-SW	Fa0/1 and Fa0/2 to both MDFs
Edge	EDGE-RTR, MPLS-ROUTER (AS 6500), cloud, DSL modem	Fiber from CORE to EDGE to MPLS-ROUTER to cloud

Endpoints: 12 laptops, 1 PC, 2 printers, 1 IP phone, 1 wireless access point with its controller.

Addressing and VLAN design

Ten routed VLANs share 128.137.232.0/22 and are split into paired subnets (30/31, 40/41, 70/71, 80/81), each with an HSRP virtual gateway, while the servers sit on 192.168.1.0/24 behind the DMZ router. DHCP is centralized: the SVIs relay to the DHCP server at 192.168.1.3, which holds one pool per VLAN and hands out the DNS server at 192.168.1.2.

VLAN	Purpose	Subnet	HSRP gateway
30	User	128.137.232.0/25	128.137.232.126
31	User	128.137.232.128/25	128.137.232.254
40	Voice	128.137.233.0/25	128.137.233.126
41	Voice	128.137.233.128/25	128.137.233.254
70	Printer	128.137.234.0/28	128.137.234.14
71	Printer	128.137.234.16/28	128.137.234.30
80	VIP	128.137.234.32/28	128.137.234.46
81	VIP	128.137.234.48/28	128.137.234.62

VLAN	Purpose	Subnet	HSRP gateway
100	Wireless	128.137.234.96/27	128.137.234.126
999	Management	128.137.234.64/27	128.137.234.94
90	Native	none	none
Servers	DNS, DHCP, web, email, TFTP, AAA, NTP	192.168.1.0/24	DMZ-RTR 192.168.1.1

Other address space:

- **Point-to-point links:** /30s from 128.137.234.160 to .183 join the DMZ router, central router, both building cores, both MDFs and the edge router.
- **GRE tunnels:** /30s from 128.137.234.188 to .211.
- **Management:** Vlan999 addresses .65 to .79 cover the server switch, routers, MDFs, access switches and the wireless switch.
- **Transit:** 10.120.100.0/22 between EDGE and the MPLS router, and 10.120.200.0/22 on its outside link.

Switching and Layer 3

- **Three tiers:** six 2960 access switches (FL1 to FL6) uplink to two 3560 Layer 3 distribution switches (BLDG20-MDF-L3 and BLDG30-MDF-L3), which uplink to their building core routers and then the central CORE-RTR/FW.
- **Dual-homed access layer:** every access switch has Fa0/1 and Fa0/2 to both MDFs (FL1 to FL3 prefer MDF20 on Fa0/1; FL4 to FL6 prefer MDF30). A copper trunk between the MDFs on Gi0/2 ties the pair together.
- **Trunks:** native VLAN 90 (unused for data) with an allowed list of 30-31, 40-41, 70-71, 80-81, 90, 100 and 999.
- **Inter-VLAN routing:** `ip routing` on both MDFs, with an SVI per VLAN and HSRP version 2 groups on each. BLDG20-MDF is active (priority 110, preempt); BLDG30-MDF is the standby (default priority, preempt).
- **Access ports:** about 20 ports per switch in VLAN 30 with voice VLAN 40; a few ports are placed in the printer, VIP, wireless and management VLANs.
- **Voice and wireless:** an IP phone on FL4 with a laptop behind it; a WLC on its own switch (WLC-SW) and an access point on FL1 Fa0/23 serving SSID ColesProduction in VLAN 100.

Routing and redundancy

Protocol	Where	Notes
OSPF 1, area 0	All routers and both MDFs	CORE router-id 1.1.1.1
EIGRP 100	CORE and EDGE side	CORE redistributes OSPF and EIGRP in both directions
eBGP	EDGE (AS 6550) to MPLS-ROUTER (AS 6500)	EDGE redistributes between EIGRP and BGP; peer 10.120.100.2
GRE	Six tunnels	DMZ-CORE, CORE-BLDG20, BLDG20-MDF20, CORE-BLDG30, BLDG30-MDF30, CORE-EDGE

Redundancy in place

- HSRP gateway pairs for every user, voice, printer, VIP and wireless VLAN.
- Every access switch reaches both distribution switches.
- Both building cores can carry traffic to the central router.

Single points of failure: the central CORE-RTR/FW, the DMZ router and the EDGE router each exist once, so a failure there isolates a whole section. Scope is intranet only, so there is no NAT or default route on EDGE yet.

Services and security

Service	Address	Notes
DNS	192.168.1.2	One A record pointing the resume site at 192.168.1.4
DHCP	192.168.1.3	10 pools; SVIs relay with `ip helper-address`
Web	192.168.1.4	HTTP only
Email	192.168.1.5	Domain colesnetwork.vet, 3 mailboxes
TFTP	192.168.1.6	
AAA	192.168.1.7	TACACS+; 13 clients enrolled

Service	Address	Notes
NTP	192.168.1.8	Server enabled with MD5 key 10

Security as built

- Servers sit behind a dedicated DMZ router, separate from the campus core.
- TACACS+ is configured on Server-RM (`aaa authentication login VTY group tacacs+`, exec authorization, console left local).
- SSH v2 is enabled on 14 of 16 devices (missing on BLDG20-CORE-RTR and EDGE-RTR) and `service password-encryption` on 9 of 16.
- SNMP read-write community on 8 devices.
- No ACLs, firewall zones, port security, PortFast or BPDU guard yet.

Testing and verification

No run-time test results (pings, failover, tunnel state) were available for this file, so the findings below come from reading the saved configuration, not from live testing.

- **GRE destination typos:** BLDG30-CORE-RTR Tunnel4 points to 128.127.234.178 (should be 128.137.234.178) and EDGE-RTR Tunnel5 points to 182.137.234.181 (should be 128.137.234.181). 10 of 12 tunnel endpoints are correct.
- **Duplicate SVI addresses:** BLDG30-MDF-L3 reuses BLDG20-MDF-L3's SVI addresses on VLANs 30, 31, 40, 41, 70, 71, 80, 81 and 100. HSRP needs a unique physical address per switch, so the pair cannot work as designed. Only Vlan999 is unique.
- **DHCP gateways:** pools hand out the SVI address as gateway rather than the HSRP virtual IP, so clients would lose their gateway on a failover. Pools also start at the network address.
- **AAA coverage:** 13 devices are enrolled on the server but only Server-RM is configured to use it.
- **Management:** FL1's management IP is a typo (129.137.234.72) and the access switches have no default gateway.
- **NTP:** the server is enabled but no device is set up as a client.
- **Hardening gaps:** SSH and password encryption are not on every device (see Services and security).

Skills and next steps

Skills shown: three-tier campus design, VLAN and subnet planning, HSRPv2 gateway redundancy, multilayer switching, OSPF/EIGRP/BGP with redistribution, GRE tunnels, DHCP relay, a wireless controller deployment, and a services segment (DNS, DHCP, web, email, TFTP, AAA, NTP).

Gaps in this build and how CollapseCore handles them

Gap in this build	Handled in CollapseCore
Duplicate SVI addresses on the second MDF	Unique SVI addresses per switch
DHCP gateways use SVI addresses	Gateways set to the HSRP virtual IPs
2 of 12 GRE endpoints wrong	All 12 endpoints validated
No port security, PortFast or BPDU guard	All three applied
No STP root priorities	Root placement set on the distribution pair
No management default gateways	Added
No NTP clients	Added
AAA only on one device	RADIUS on 9 devices
No firewall	Zone-based firewall with 3 zones
No HSRP interface tracking	Tracking on the uplink to the firewall

Next steps: fix the GRE typos and duplicate SVIs, apply the same hardening here, then run and record failover and reachability tests.